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Welcome to TinyML

Part 1 | Introduction to TinyML & ML Basics

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Part 3 | Data Acquisition

Part 3.1 | Build a data acquisition system

Part 5 | Feature engineering

Part 5.1 | Implement feature calculation on your device

Part 6 | Introduction to Machine Learning

Part 7 | Introduction to EM Learn, ML on Microcontr

Part 7.1 | Implement a ML model trained on your fei

Part 8 | Introduction to Deep Learning

Part 8.1 | Introduction to Convolutions

Part 9 | Introduction to AiFES, Deep Learning on

Part 10 | Optimisation techniques, Quantisation

Part 11 | Optimisation techniques, Pruning

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Semester/Exam Project

Unclaiming your device - HUSK

Welcome to TinyML

Welcome to the TinyML course

The Tiny Machine Learning (TinyML) course is an introductory course at the intersection of Machine Learning, Signal Processing and Embedded IoT systems.



TinyML has formally been around since 2019, when the TinyML foundation was established, and just recently it changed its name to edgeaifoundation.

Performing analysis and signal processing close to the origin of the data is nothing new, but the possibilities with modern hardware, such as 32bit low power microcontroller allows us to perform both signal processing and machine learning on compact models rather efficient.

It allows the usage of even small edge devices like Cortex M CPU's with kilobytes of RAM to perform tasks like analytics and anomaly detection, on devices consuming only a few milliwatts, often an order of magnitude or more below the energy cost of transmitting raw data.

This course:

In the course you will be *presented* to:

- TinyML applications and use cases
- Algorithms suited for optimization and deployment on a 32bit MCU
- Tiny ML Frameworks, tools and techniques
- The importance of good data sets
- Data preprocessing and introductory feature engineering
- Low power design techniques

After the course you will have *achieved* the folowing:

- Familiarity with one or more toolchains for TinyML
- Hand on experience with model design, training and deployment on an embedded device
- Model optimisation for memory and power constrained devices
- Data collection and labeling

The course requires active participation and collaboration and will be a mix of reading and discussing academic literature, theory presentations and hands-on lab exercises.

We will work with one or more suited embedded platforms for data collection and model deployment, ideally on participants' own use cases.

During the course you will spend a significant portion of time working on a semester projects, that is presented at the exam.